

ELITE IGCSE ACADEMY

Student Past Paper Solutions

May 2025 4WM2HR Worked Solutions

May2025 | 4WM2HR | Unit 2

31 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 5**TOPIC: **Statistics Toolkit**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

- 1 The table shows information about the weights, in kilograms, of 50 dogs.

Weight (w kg)	Frequency
$4 < w \leq 8$	5
$8 < w \leq 12$	17
$12 < w \leq 16$	15
$16 < w \leq 20$	9
$20 < w \leq 24$	4

- (a) Write down the modal class.

..... kg
(1)

- (b) Work out an estimate for the mean weight of the 50 dogs.

..... kg
(4)

(Total for Question 1 is 5 marks)

PAST PAPER SOLUTION**Q#: 1****Marks: 5****TOPIC: Statistics Toolkit**

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WORKED SOLUTION*Dr. Eslam Ahmed***Required:***a* Write down the modal class. (b) Estimate the mean weight of the 50 dogs.**Part (a)**

The **modal class** is the interval with the highest frequency. From the table, the highest frequency is **17**, which belongs to:

FINAL ANSWERModal class: $8 < w \leq 12$ kg**Part (b)****Step:**

Find the midpoint of each class interval:

Step:

Apply the estimated mean formula:

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{660}{50} = 13.2 \text{ kg}$$

FINAL ANSWER*Estimated mean weight = 13.2kg*

PAST PAPER QUESTION

Q#: 2

Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

- 2 Write 1400 as a product of powers of its prime factors.
Show your working clearly.

.....
(Total for Question 2 is 3 marks)

Dr. Es

PAST PAPER SOLUTION**Q#: 2****Marks: 3****TOPIC: Prime Factors, HCF & LCM**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***Required:**

Write 1400 as a product of powers of its prime factors.

Step:

Divide repeatedly by the smallest prime factor each time:

$$1400 \xrightarrow{\div 2} 700 \xrightarrow{\div 2} 350 \xrightarrow{\div 2} 175 \xrightarrow{\div 5} 35 \xrightarrow{\div 5} 7 \xrightarrow{\div 7} 1$$

Step:

Collect the prime factors:

$$1400 = \underbrace{2 \times 2 \times 2}_{2^3} \times \underbrace{5 \times 5}_{5^2} \times 7$$

FINAL ANSWER

$$1400 = 2^3 \times 5^2 \times 7$$

PAST PAPER QUESTION**Q#: 3****Marks: 3****TOPIC: Simultaneous Equations**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

3 Solve the simultaneous equations

$$3x + 2y = 10$$

$$3x - 4y = 16$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 3****TOPIC: Simultaneous Equations**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***Required:**Solve: $3x + 2y = 10$ and $3x - 4y = 16$ algebraically.**Step:**

Label the equations:

$$3x + 2y = 10 \quad (1)$$

$$3x - 4y = 16 \quad (2)$$

Step:Subtract (2) from (1) to eliminate x :

$$(1) - (2) : \quad 6y = -6 \implies \boxed{y = -1}$$

Step:Substitute $y = -1$ into (1):

$$3x + 2(-1) = 10 \implies 3x = 12 \implies \boxed{x = 4}$$

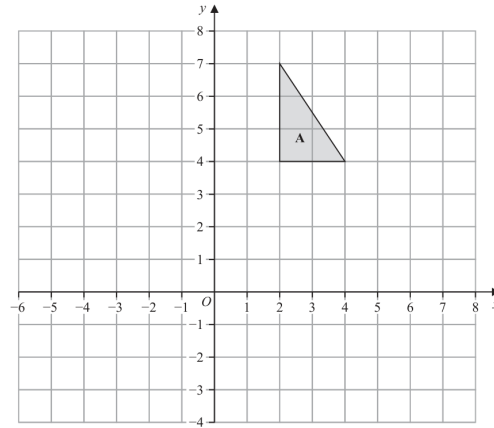
Step:Check in equation (2): $3(4) - 4(-1) = 12 + 4 = 16$ **FINAL ANSWER**

$$x = 4, y = -1$$

PAST PAPER QUESTION**Q#: 4****Marks: 4****TOPIC: Transformations**

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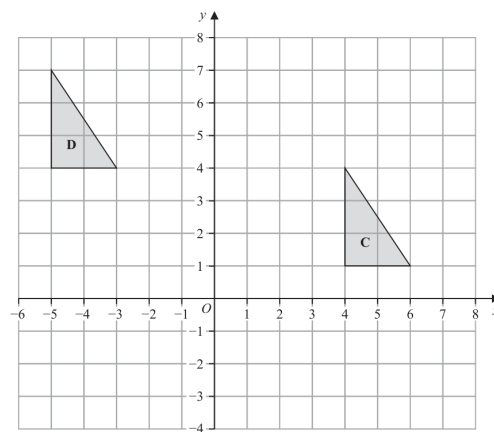
4



- (a) On the grid above, rotate triangle A 90° anticlockwise about (1, 2)

Label your triangle B

(2)



- (b) Describe fully the single transformation that maps triangle C onto triangle D

(2)

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION**Q#: 4** **Marks: 4****TOPIC: Transformations**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***Required:**

a Rotate triangle A 90° anticlockwise about (1,2); label triangle B. (b) Describe fully the transformation mapping C onto D.

Part (a)**Step:**

The rule for 90° anticlockwise rotation about (a,b) :

$$(x, y) \rightarrow (a - (y-b), b + (x-a))$$

About (1,2): $(x,y) \rightarrow (3-y, x+1)$

Step:

Triangle A has vertices $A_1(2,7)$, $A_2(4,5)$, $A_3(2,5)$. Apply the rule:

$$(2,7) \rightarrow (-4,3), (4,5) \rightarrow (-2,5), (2,5) \rightarrow (-2,3)$$

FINAL ANSWER

Triangle B has vertices $(-4,3)$, $(-2,5)$, $(-2,3)$

Part (b)**Step:**

Identify corresponding vertices of C and D from the diagram and compute the vector:

$$(4, 4) \rightarrow (-5, 7) : \quad \vec{v} = \begin{pmatrix} -9 \\ 3 \end{pmatrix}$$

Check: $(6,1) \rightarrow (-3,4)$ gives the same vector $\begin{pmatrix} -9 \\ 3 \end{pmatrix}$

FINAL ANSWER

Translation by vector $\begin{pmatrix} -9 \\ 3 \end{pmatrix}$

PAST PAPER QUESTION**Q#: 5****Marks: 5****TOPIC: Percentages**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

- 5** Joshua is going to cover a floor with tiles for a customer.
The area of the floor is 45 m^2

Joshua buys one box of tiles for each 1.5 m^2 of floor area.
Each box of tiles costs £64

Joshua also buys 5 bags of tile adhesive.
Each bag of tile adhesive costs £12

Joshua charges the customer £3000

Work out his percentage profit.
Give your answer correct to one decimal place.

.....%

(Total for Question 5 is 5 marks)

PAST PAPER SOLUTION**Q#: 5** **Marks: 5**TOPIC: **Percentages**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***Required:**

Calculate Joshua's percentage profit on the flooring job. Give answer to 1 d.p.

Step:

Number of boxes of tiles needed:

$$\frac{45 \text{ m}^2}{1.5 \text{ m}^2/\text{box}} = 30 \text{ boxes}$$

Step:

Cost of tiles:

$$30 \times £64 = £1920$$

Step:

Cost of adhesive:

$$5 \times £12 = £60$$

Step:

Total cost (cost price):

$$£1920 + £60 = £1980$$

Step:

Profit and percentage profit:

$$\text{Profit} = £3000 - £1980 = £1020$$

$$\% \text{ Profit} = \frac{1020}{1980} \times 100 = 51.515 \dots \%$$

FINAL ANSWER

Percentage profit = 51.5%

Note:

Always divide profit by the **cost price** (£1980), not the selling price (£3000). Dividing by £3000 gives 34.0% – a common but wrong answer.

PAST PAPER QUESTION**Q#: 6****Marks: 3****TOPIC: Powers, Roots & Standard Form**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

6 (a) Write 82 000 000 in standard form.

.....
(1)

(b) Write 5.8×10^{-6} as an ordinary number.

.....
(1)

(c) Write 160×10^{300} in standard form.

.....
(1)

(Total for Question 6 is 3 marks)

PAST PAPER SOLUTION**Q#: 6****Marks: 3****TOPIC: Powers, Roots & Standard Form**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***Required:**

a Write 82 000 000 in standard form. (b) Write 5.8×10^{-6} as an ordinary number. (c) Write 160×10^{300} in standard form.

Part (a)

$$82000000 = 8.2 \times 10^7$$

FINAL ANSWER

$$8.2 \times 10^7$$

Part (b)

Move the decimal 6 places **left**:

$$5.8 \times 10^{-6} = 0.0000058$$

FINAL ANSWER

$$0.0000058$$

Part (c)

$$160 \times 10^{300} = 1.6 \times 10^1 \times 10^{300} = 1.6 \times 10^{301}$$

FINAL ANSWER

$$1.6 \times 10^{301}$$

PAST PAPER QUESTION

Q#: 7

Marks: 3

TOPIC: **Statistics Toolkit**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

7 The mean of 7 numbers is 60

The mean of 3 of the numbers is 46

Work out the mean of the other 4 numbers.

.....
(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION**Q#: 7****Marks: 3**TOPIC: **Statistics Toolkit**

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WORKED SOLUTION

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Required:*Mean of 7 numbers = 60; mean of 3 of them = 46. Find the mean of the other 4.***Step:**

Sum of all 7 numbers:

$$7 \times 60 = 420$$

Step:

Sum of the 3 numbers:

$$3 \times 46 = 138$$

Step:

Sum of the remaining 4:

$$420 - 138 = 282$$

Step:

Mean of the 4 numbers:

$$\frac{282}{4} = 70.5$$

FINAL ANSWER*Mean of the other 4 numbers = 70.5*

PAST PAPER QUESTION**Q#: 8****Marks: 3**TOPIC: **Percentages**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

- 8 In a sale, normal prices are reduced by 15%
- The sale price of a dishwasher is 612 Swiss francs.
- Work out the normal price of the dishwasher.

..... Swiss francs

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 3**TOPIC: **Percentages**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Required:

Sale price (after 15% reduction) = 612 Swiss francs. Find the normal price.

Step:

A 15% reduction means the sale price is 85% of the normal price:

$$612 = 0.85 \times \text{Normal price}$$

Step:

Solve for the normal price:

$$\text{Normal price} = \frac{612}{0.85} = 720$$

FINAL ANSWER*Normal price = 720 Swiss francs***Note:**

Never calculate 15% of 612 and add it on – that gives 703.80 (wrong!). Always divide by the decimal multiplier.

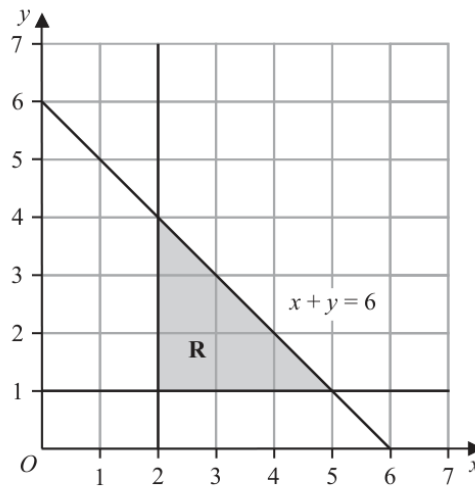
PAST PAPER QUESTION**Q#: 9****Marks: 6****TOPIC: Graphing Inequalities**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

- 9 (a) Solve the inequality $3x - 22 > 8x - 40$
Show clear algebraic working.

.....
(3)

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



- (b) Write down the three inequalities that define the region **R**

.....
.....
.....
(3)

(Total for Question 9 is 6 marks)

PAST PAPER SOLUTION**Q#: 9****Marks: 6****TOPIC: Graphing Inequalities**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Required:a Solve $3x - 22 > 8x - 40$ algebraically. (b) Write the three inequalities defining region R.**Part (a)****Step:**Move all x terms to the right and constants to the left:

$$3x - 22 > 8x - 40 \implies -22 + 40 > 8x - 3x \implies 18 > 5x$$

Step:

Divide both sides by 5:

$$x < \frac{18}{5} = 3.6$$

FINAL ANSWER

$$x < 3.6$$

Part (b)**Step:**

Identify the three boundary lines and their regions using the graph. The shaded region R is bounded by $x = 2$, $y = 1$, and $x + y = 6$.

FINAL ANSWER

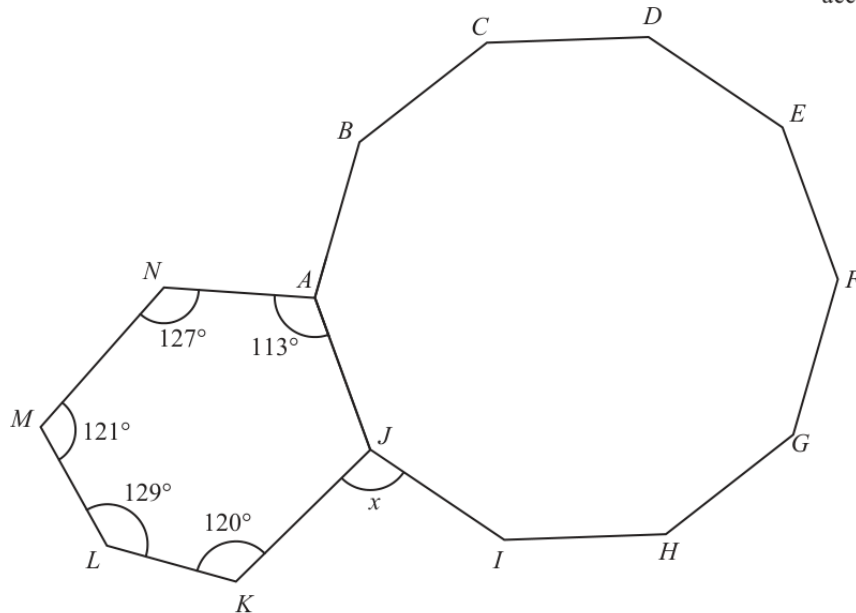
$$x \geq 2, y \geq 1, x + y \leq 6$$

PAST PAPER QUESTION**Q#: 10****Marks: 5****TOPIC: Angles in Polygons & Parallel Lines**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

- 10** The diagram shows a regular 10-sided polygon $ABCDEFGHIJ$ and a hexagon $AJKLMN$

Diagram **NOT**
accurately drawn



Work out the size of the angle marked x
Show your working clearly.

.....
(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 5****TOPIC: Angles in Polygons & Parallel Lines**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***Required:**Find angle x in the figure combining a regular 10-gon (ABCDEFGHJIJ) and hexagon AJKLMN.**Step:**

Interior angle of a regular 10-sided polygon:

$$\text{Interior angle} = \frac{(10 - 2) \times 180^\circ}{10} = \frac{1440^\circ}{10} = 144^\circ$$

Step:Find the interior angle of the hexagon at vertex A. At point A: decagon angle + 113° + hexagon angle at A = 360° :

$$\angle x = 360^\circ - 144^\circ - 110^\circ = 106^\circ$$

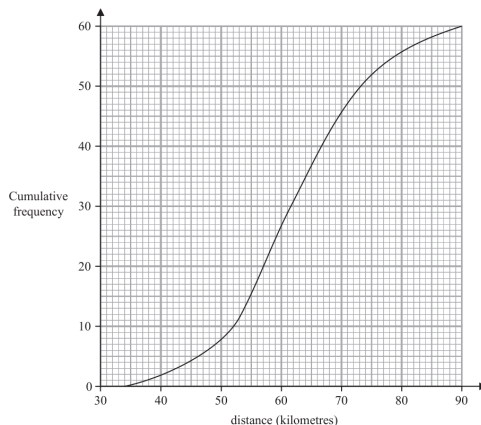
FINAL ANSWER

$$x = 106^\circ$$

PAST PAPER QUESTION**Q#: 11****Marks: 7****TOPIC: Cumulative Frequency Diagrams**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

- 11 The cumulative frequency graph shows information about the distance, in kilometres, Ehrick drives each day for 60 days.



- (a) Use the graph to find an estimate for the median of the distances.

_____ kilometres
(1)

- (b) Use the graph to find an estimate for the interquartile range of the distances.

_____ kilometres
(2)

- (c) Work out the percentage of days that Ehrick drives for more than 75 kilometres.
Give your answer correct to one decimal place.
Show your working clearly.

_____ %
(3)

Tomas and Alison record the number of kilometres that they drive each day for 60 days.
The table gives information about their results.

	Interquartile range
Tomas	28 kilometres
Alison	15 kilometres

- (d) Which of Tomas or Alison is more consistent in the number of kilometres that they drive.
Give a reason for your answer.

(1)

(Total for Question 11 is 7 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 7****TOPIC: Cumulative Frequency Diagrams**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Required:

a Median(b) IQR(c) % days > 75 km(d) More consistent?

Part (a)**Step:**Median at the $\frac{60}{2} = 30$ th value – read from graph at cf = 30:**FINAL ANSWER**Median ≈ 62 km**Part (b)****Step:** $Q_1 \text{ at } cf = 15, Q_3 \text{ at } cf = 45.$ Read from graph : $Q_1 \approx 56$ km, $Q_3 \approx 74$ kmIQR = $Q_3 - Q_1 = 74 - 56 = 18$ km**FINAL ANSWER**IQR ≈ 18 km**Note:**Graph-reading answers are accepted within ± 2 km of the model answer.**Part (c)****Step:**

From the graph, read the cumulative frequency at d = 75 km :

cf at 75 km ≈ 52 **Step:**

Number of days with distance > 75 km:

 $60 - 52 = 8$ days**Step:**

Percentage:

$$\frac{8}{60} \times 100 = 13.3\%$$

FINAL ANSWER

13.3% of days Ehrick drives more than 75 km

Part (d)**FINAL ANSWER**

Alison is more consistent – her IQR (15 km) is smaller than Tomas's (28 km), meaning less variability in daily distances.

PAST PAPER QUESTION**Q#: 12****Marks: 3****TOPIC: Compound Interest & Depreciation**

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12 Osman buys a car for \$16 000

The car depreciates at a rate of 12% each year for the first 2 years.

In the third year, the car depreciates at a rate of $x\%$

At the end of 3 years, the value of the car is \$11 461.12

Work out the value of x

.....

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION**Q#: 12****Marks: 3****TOPIC: Compound Interest & Depreciation**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***Required:**

Car bought for \$16 000; depreciates 12%/yr for 2 yrs, then $x\%$ /yr in year 3. Value after 3 years = \$11 461.12. Find x .

Step:

Value after 2 years at 12% depreciation:

$$V_2 = 16000 \times (0.88)^2 = 16000 \times 0.7744 = \$12\,390.40$$

Step:

Apply year-3 depreciation and set equal to final value:

$$12390.40 \times \left(1 - \frac{x}{100}\right) = 11461.12$$

Step:

Solve:

$$1 - \frac{x}{100} = \frac{11461.12}{12390.40} = 0.925 \implies \frac{x}{100} = 0.075 \implies x = 7.5$$

FINAL ANSWER

$$x = 7.5$$

PAST PAPER QUESTION**Q#: 13****Marks: 3****TOPIC: Powers, Roots & Standard Form**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

13 The distance from Earth to Andromeda Galaxy is 1.6×10^{11} astronomical units.

- 1 astronomical unit = 1.5×10^8 km
- 1 light year = 9.5×10^{12} km

Work out the distance, in light years, from Earth to Andromeda Galaxy.

Give your answer in standard form, correct to 2 significant figures.

..... light years

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION**Q#: 13** **Marks: 3****TOPIC: Powers, Roots & Standard Form**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***Required:**Distance to Andromeda: 1.6×10^{11} AU. Convert to light years (2 s.f., standard form).**Step:**Convert AU $\rightarrow$ km:

$$1.6 \times 10^{11} \times 1.5 \times 10^8 = (1.6 \times 1.5) \times 10^{19} = 2.4 \times 10^{19} \text{ km}$$

Step:Convert km $\rightarrow$ light years:

$$\frac{2.4 \times 10^{19}}{9.5 \times 10^{12}} = \frac{2.4}{9.5} \times 10^7 = 0.2526 \dots \times 10^7$$

Step:

Write in standard form to 2 s.f.:

$$= 2.526 \dots \times 10^6 \approx 2.5 \times 10^6$$

FINAL ANSWER

$$\text{Distance} = 2.5 \times 10^6 \text{ lightyears}$$

PAST PAPER QUESTION**Q#: 14****Marks: 4****TOPIC: Rearranging Formulas**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

14 Make k the subject of $p = \frac{8k^2 + 5}{7 - 3k^2}$

.....
(Total for Question 14 is 4 marks)

Dr. Eslam

PAST PAPER SOLUTION**Q#: 14****Marks: 4****TOPIC: Rearranging Formulas**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***Required:**

Make k the subject of $p = \frac{8k^2 + 5}{7 - 3k^2}$.

Step:

Multiply both sides by $(7 - 3k^2)$:

$$p(7 - 3k^2) = 8k^2 + 5$$

Step:

Expand:

$$7p - 3pk^2 = 8k^2 + 5$$

Step:

Collect all k^2 terms on one side:

$$7p - 5 = 8k^2 + 3pk^2$$

Step:

Factorise and solve for k :

$$7p - 5 = k^2(3p + 8) \implies k^2 = \frac{7p - 5}{3p + 8}$$

$$k = \pm \sqrt{\frac{7p - 5}{3p + 8}}$$

FINAL ANSWER

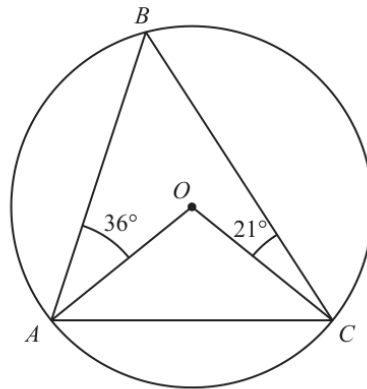
$$k = \pm \sqrt{\frac{7p - 5}{3p + 8}}$$

Note:

Include both $\pm$ roots unless the context specifies the sign of k .

PAST PAPER QUESTION**Q#: 15****Marks: 3****TOPIC: Circle Theorems**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

15 A , B and C are points on a circle, centre O Diagram **NOT**
accurately drawnAngle $BAO = 36^\circ$ Angle $BCO = 21^\circ$ Work out the size of angle ACO

.....
 (Total for Question 15 is 3 marks)

PAST PAPER SOLUTION**Q#: 15** **Marks: 3**TOPIC: **Circle Theorems**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Required:A, B, C on circle centre O. $\angle^*BAO = 36^\circ$, $\angle BCO = 21^\circ$. Find $\angle ACO^*$.**Step:** $\triangle^*OAB$ is isosceles ($OA^* = OB^*$ = radius):

$$\angle^*OAB = \angle OBA = 36^\circ \implies \angle AOB^* = 180^\circ - 36^\circ - 36^\circ = 108^\circ$$

Step: $\triangle^*OBC$ is isosceles ($OB^* = OC^*$):

$$\angle^*OBC = \angle OCB = 21^\circ \implies \angle BOC^* = 180^\circ - 21^\circ - 21^\circ = 138^\circ$$

Step:

Angles at centre O (on the same side):

$$\angle^*AOC^* = 360^\circ - 108^\circ - 138^\circ = 114^\circ$$

Step: $\triangle^*OAC$ is isosceles ($OA^* = OC^*$):

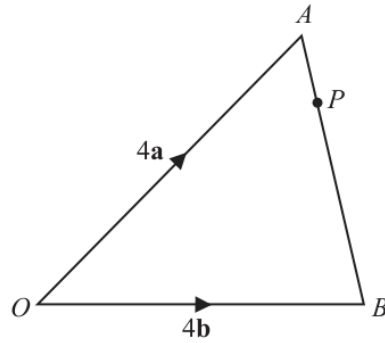
$$\angle OAC + \angle OCA = 180^\circ - 114^\circ = 66^\circ \implies \angle ACO = \frac{66^\circ}{2} = 33^\circ$$

FINAL ANSWER

$$\angle^*ACO^* = 33^\circ$$

PAST PAPER QUESTION**Q#: 16****Marks: 3**TOPIC: **Vectors**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

16 OAB is a triangle.Diagram **NOT**
accurately drawn

$$\vec{OA} = 4\mathbf{a}$$

$$\vec{OB} = 4\mathbf{b}$$

 P is the point on AB such that $AP : PB = 1 : 3$ (a) Write down $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
(1)(b) Express $\vec{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form......
(2)**(Total for Question 16 is 3 marks)**

PAST PAPER SOLUTION**Q#: 16** **Marks: 3**TOPIC: **Vectors**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Required: $\vec{OA} = 4\mathbf{a}$, $\vec{OB} = 4\mathbf{b}$, $AP:PB = 1:3$. (a) Find $\vec{AB}$. (b) Express $\vec{OP}$ in simplest form.**Part (a)**

$$\vec{AB} = \vec{AO} + \vec{OB} = -4\mathbf{a} + 4\mathbf{b}$$

FINAL ANSWER

$$\vec{AB} = 4\mathbf{b} - 4\mathbf{a}$$

Part (b)**Step:** $AP:PB = 1:3$ means $\vec{AP} = \frac{1}{4}\vec{AB}$:

$$\vec{AP} = \frac{1}{4}(4\mathbf{b} - 4\mathbf{a}) = \mathbf{b} - \mathbf{a}$$

Step:Use $\vec{OP} = \vec{OA} + \vec{AP}$:

$$\vec{OP} = 4\mathbf{a} + (\mathbf{b} - \mathbf{a}) = 3\mathbf{a} + \mathbf{b}$$

FINAL ANSWER

$$\vec{OP} = 3\mathbf{a} + \mathbf{b}$$

PAST PAPER QUESTION**Q#: 17****Marks: 4****TOPIC: Functions**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

17 The function g is defined as

$$g(x) = \frac{3x}{x-2}$$

(a) State the value of x that cannot be included in any domain of g
(1)(b) Express the inverse function g^{-1} in the form $g^{-1}(x) = \dots$

$$g^{-1}(x) = \dots\dots\dots$$

(3)

(Total for Question 17 is 4 marks)

PAST PAPER SOLUTION**Q#: 17****Marks: 4****TOPIC: Functions**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Required: $g(x) = \frac{3x}{x-2}$. (a) Value excluded from domain. (b) Find $g^{-1}(x)$.**Part (a)**The denominator must not equal zero: $x - 2 \neq 0$.**FINAL ANSWER** $x = 2$ cannot be in any domain of g **Part (b)****Step:**Set $y = g(x)$ and swap $x \leftrightarrow y$:

$$x = \frac{3y}{y-2}$$

Step:Multiply both sides by $(y-2)$:

$$x(y-2) = 3y \implies xy - 2x = 3y$$

Step:Collect y terms and factorise:

$$xy - 3y = 2x \implies y(x-3) = 2x$$

Step:Solve for y :

$$y = \frac{2x}{x-3}$$

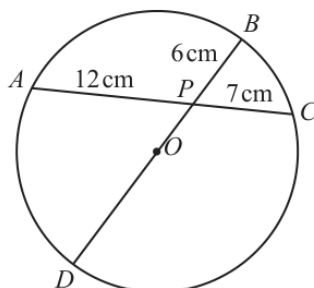
FINAL ANSWER

$$g^{-1}(x) = \frac{2x}{x-3}$$

Note: $g^{-1}(x)$ is undefined at $x = 3$, which is the range restriction of $g(x)$ (horizontal asymptote).

PAST PAPER QUESTION**Q#: 18****Marks: 3****TOPIC: Circle Theorems**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

18Diagram **NOT**
accurately drawn A, B, C and D are points on a circle, centre O DB is a diameter of the circle. AC and DB intersect at the point P

$$AP = 12 \text{ cm} \quad PC = 7 \text{ cm} \quad PB = 6 \text{ cm}$$

Work out the area of the circle.

Give your answer correct to 3 significant figures.

..... cm^2 **(Total for Question 18 is 3 marks)**

PAST PAPER SOLUTION**Q#: 18****Marks: 3****TOPIC: Circle Theorems**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Required:

DB is a diameter; chords AC and DB intersect at P . $AP = 12\text{ cm}$, $PC = 7\text{ cm}$, $PB = 6\text{ cm}$. Find area of circle (3s.f.).

Step:

Apply the **Intersecting Chords Theorem** : $AP \times PC = BP \times PD$

$$12 \times 7 = 6 \times PD \implies PD = \frac{84}{6} = 14 \text{ cm}$$

Step:

Diameter DB :

$$DB = PD + PB = 14 + 6 = 20 \text{ cm} \implies r = 10 \text{ cm}$$

Step:

Area of circle:

$$\text{Area} = \pi r^2 = 100\pi \approx 314.159 \dots \text{ cm}^2$$

FINAL ANSWER

$$\text{Area of circle} = 314 \text{ cm}^2 \text{ (3s.f.)}$$

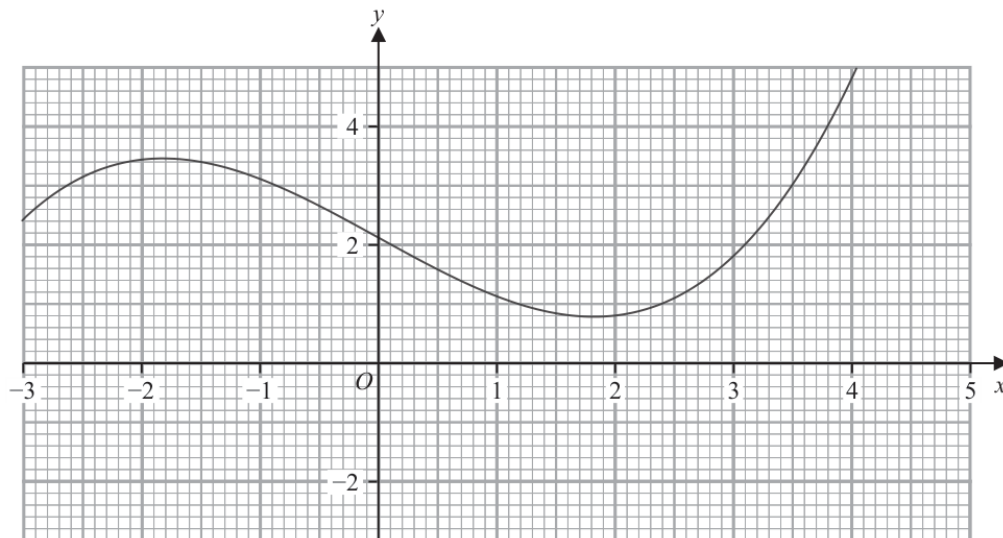
Note:

Intersecting Chords Theorem : if two chords cross inside a circle at P , then $AP \times PC = BP \times PD$.

PAST PAPER QUESTION**Q#: 19****Marks: 3****TOPIC: Estimating Gradients**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

19 The diagram shows the graph of $y = f(x)$



Find an estimate for the gradient of the graph at the point where $x = 3$
Show your working clearly.

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION**Q#: 19** **Marks: 3****TOPIC: Estimating Gradients**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Required:*Estimate the gradient of $y = f(x)$ at the point where $x = 3$.***Step:**

Draw a tangent to the curve at $x = 3$. From the graph, a suitable tangent passes through approximately (1, -2) and (5, 6):

Step:

Calculate gradient of the tangent:

$$\text{Gradient} = \frac{\Delta y}{\Delta x} = \frac{6 - (-2)}{5 - 1} = \frac{8}{4} = 2$$

FINAL ANSWER

Gradient ≈ 2 at $x = 3$

Note:

Answers in the range 1 to 3 are typically accepted. Draw the tangent triangle as large as possible for the most accurate estimate.

PAST PAPER QUESTION**Q#: 20****Marks: 3**TOPIC: **Solving Inequalities**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

20 Solve the inequality $2x^2 - 7x - 15 > 0$

Show clear algebraic working.

.....

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 3**TOPIC: **Solving Inequalities**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Required:Solve $2x^2 - 7x - 15 > 0$. Show clear algebraic working.**Step:**

Factorise the quadratic:

$$2x^2 - 7x - 15 = (2x+3)(x-5)$$

$$\text{Check: } (2x+3)(x-5) = 2x^2 - 10x + 3x - 15 = 2x^2 - 7x - 15$$

Step:

Roots: set each factor to zero:

$$2x + 3 = 0 \implies x = -\frac{3}{2}; \quad x - 5 = 0 \implies x = 5$$

Step:

Since the coefficient of x^2 is $+2 > 0$, the parabola opens **upward**. The expression is > 0 *outside* the roots:

FINAL ANSWER

$$x < -\frac{3}{2} \text{ or } x > 5$$

PAST PAPER QUESTION**Q#: 20****Marks: 3**TOPIC: **Solving Inequalities**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

20 Solve the inequality $2x^2 - 7x - 15 > 0$

Show clear algebraic working.

.....

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 3****TOPIC: Solving Inequalities**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***Required:**Solve $2x^2 - 7x - 15 > 0$. Show clear algebraic working.**Step:**

Factorise the quadratic:

$$2x^2 - 7x - 15 = (2x+3)(x-5)$$

$$\text{Check: } (2x+3)(x-5) = 2x^2 - 10x + 3x - 15 = 2x^2 - 7x - 15$$

Step:

Roots: set each factor to zero:

$$2x + 3 = 0 \implies x = -\frac{3}{2}; \quad x - 5 = 0 \implies x = 5$$

Step:

Since the coefficient of x^2 is $+2 > 0$, the parabola opens **upward**. The expression is > 0 *outside* the roots:

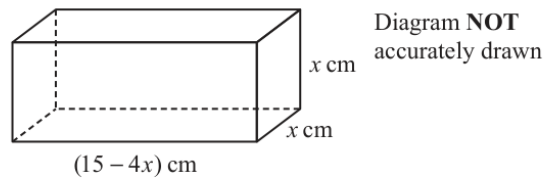
FINAL ANSWER

$$x < -\frac{3}{2} \text{ or } x > 5$$

PAST PAPER QUESTION**Q#: 21****Marks: 5**TOPIC: **Volume & Surface Area**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

- 21** The diagram shows a solid cuboid.



The volume of the cuboid is $V \text{ cm}^3$

Find the maximum value of V

.....
(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 5****TOPIC: Volume & Surface Area**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Required:

Cuboid has a square cross-section of side x cm and length $(15-4x)$ cm. Find the maximum value of the volume V .

Step:

Write V as a function of x :

$$V = x \cdot x \cdot (15-4x) = x^2(15-4x) = 15x^2 - 4x^3$$

Step:

Differentiate:

$$\frac{dV}{dx} = 30x - 12x^2$$

Step:

Set $\frac{dV}{dx} = 0$:

$$30x - 12x^2 = 0 \implies 6x(5 - 2x) = 0 \implies x = 0 \text{ or } x = \frac{5}{2} = 2.5$$

($x = 0$ gives zero volume, so the meaningful solution is $x = 2.5$.)

Step:

Second derivative test to confirm maximum:

$$\frac{d^2V}{dx^2} = 30 - 24x \Bigg|_{x=2.5} = 30 - 60 = -30 < 0 \quad \checkmark \text{ (maximum)}$$

Step:

Calculate $V_{\max}$:

$$V_{\max} = (2.5)^2 \times (15-4 \times 2.5) = 6.25 \times 5 = 31.25 \text{ cm}^3$$

FINAL ANSWER

$$\text{Maximum volume } V = 31.25 \text{ cm}^3$$

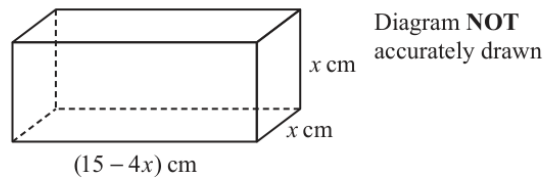
Note:

Calculus optimisation: always verify the nature of the stationary point using $\frac{d^2V}{dx^2}$. Also check $x > 0$ and $15 - 4x > 0$ (i.e. $x < 3.75$) to ensure a valid cuboid.

PAST PAPER QUESTION**Q#: 21****Marks: 5**TOPIC: **Volume & Surface Area**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

- 21** The diagram shows a solid cuboid.



The volume of the cuboid is $V \text{ cm}^3$

Find the maximum value of V

.....
(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 5****TOPIC: Volume & Surface Area**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Required:

Cuboid has a square cross-section of side x cm and length $(15-4x)$ cm. Find the maximum value of the volume V .

Step:

Write V as a function of x :

$$V = x \cdot x \cdot (15-4x) = x^2(15-4x) = 15x^2 - 4x^3$$

Step:

Differentiate:

$$\frac{dV}{dx} = 30x - 12x^2$$

Step:

Set $\frac{dV}{dx} = 0$:

$$30x - 12x^2 = 0 \implies 6x(5 - 2x) = 0 \implies x = 0 \text{ or } x = \frac{5}{2} = 2.5$$

($x = 0$ gives zero volume, so the meaningful solution is $x = 2.5$.)

Step:

Second derivative test to confirm maximum:

$$\frac{d^2V}{dx^2} = 30 - 24x \Bigg|_{x=2.5} = 30 - 60 = -30 < 0 \quad \checkmark \text{ (maximum)}$$

Step:

Calculate $V_{\max}$:

$$V_{\max} = (2.5)^2 \times (15-4 \times 2.5) = 6.25 \times 5 = 31.25 \text{ cm}^3$$

FINAL ANSWER

$$\text{Maximum volume } V = 31.25 \text{ cm}^3$$

Note:

Calculus optimisation: always verify the nature of the stationary point using $\frac{d^2V}{dx^2}$. Also check $x > 0$ and $15 - 4x > 0$ (i.e. $x < 3.75$) to ensure a valid cuboid.

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

22 Solve the simultaneous equations

$$y^2 + 5y + x^2 = 12$$

$$x = y + 5$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Required:

$$\text{Solve } y^2 + 5y + x^2 = 12 \text{ and } x = y + 5.$$

Step:

Substitute $x = y + 5$ into the first equation :

$$y^2 + 5y + (y + 5)^2 = 12$$

Step:

$$\text{Expand } (y + 5)^2 = y^2 + 10y + 25 :$$

$$y^2 + 5y + y^2 + 10y + 25 = 12 \implies 2y^2 + 15y + 13 = 0$$

Step:

Factorise:

$$(2y + 13)(y + 1) = 0 \implies y = -\frac{13}{2} \text{ or } y = -1$$

Step:

Find corresponding x values using $x = y + 5$:

FINAL ANSWER

$$(x, y) = (4, -1) \text{ and } \left(-\frac{3}{2}, -\frac{13}{2}\right)$$

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

22 Solve the simultaneous equations

$$y^2 + 5y + x^2 = 12$$

$$x = y + 5$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Required:

$$\text{Solve } y^2 + 5y + x^2 = 12 \text{ and } x = y + 5.$$

Step:

Substitute $x = y + 5$ into the first equation :

$$y^2 + 5y + (y + 5)^2 = 12$$

Step:

$$\text{Expand } (y + 5)^2 = y^2 + 10y + 25 :$$

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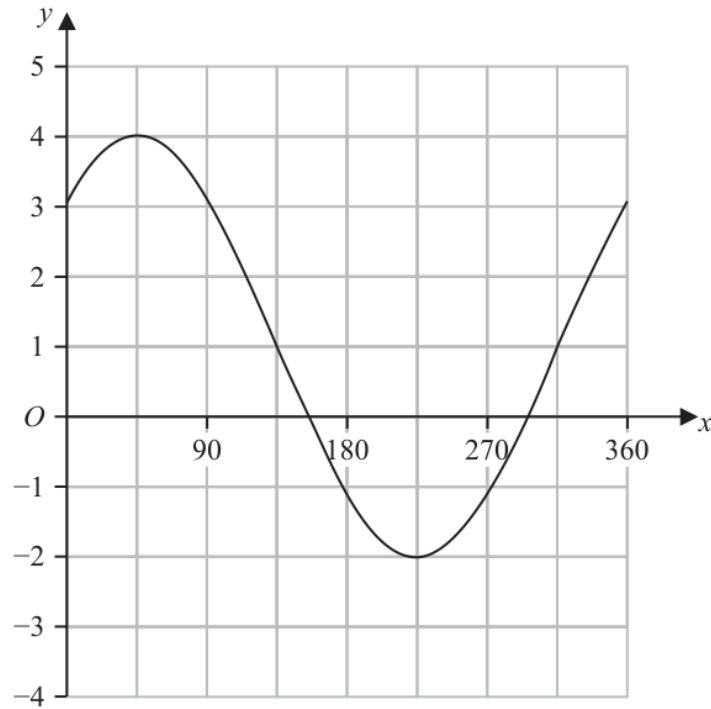
FINAL ANSWER

$$(x, y) = (4, -1) \text{ and } \left(-\frac{3}{2}, -\frac{13}{2}\right)$$

PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Graphs of Functions**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

23 The graph of $y = a\sin(x + b)^\circ + c$ for $0 \leq x \leq 360$ is drawn on the grid below.



Find a suitable value for a , for b and for c

$a = \dots\dots\dots$

$b = \dots\dots\dots$

$c = \dots\dots\dots$

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 3****TOPIC: Graphs of Functions**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***Required:**

The graph of $y = a \sin (x+b)^{\circ} + c$ for $0 \leq x \leq 360$ is shown. Find suitable values for a , b , and c .

Step:

Read the maximum and minimum values from the graph to find the **amplitude** a and **vertical shift** c :

$$a = \frac{\max - \min}{2} = \frac{4 - (-2)}{2} = 3 \quad c = \frac{\max + \min}{2} = \frac{4 + (-2)}{2} = 1$$

Step:

Find the **phase shift** b . The standard sin has its peak at $x = 90^{\circ}$. From the graph, the maximum occurs at $x \approx 45^{\circ}$:

$$x + b = 90^{\circ} \implies 45^{\circ} + b = 90^{\circ} \implies b = 45^{\circ}$$

Step:

Verify at $x = 0$: $y = 4 \sin(0 + 45)^{\circ} = 4 \times \frac{\sqrt{2}}{2} \approx 2.83$ – consistent with graph

FINAL ANSWER

$$a = 3, b = 45, c = 1$$

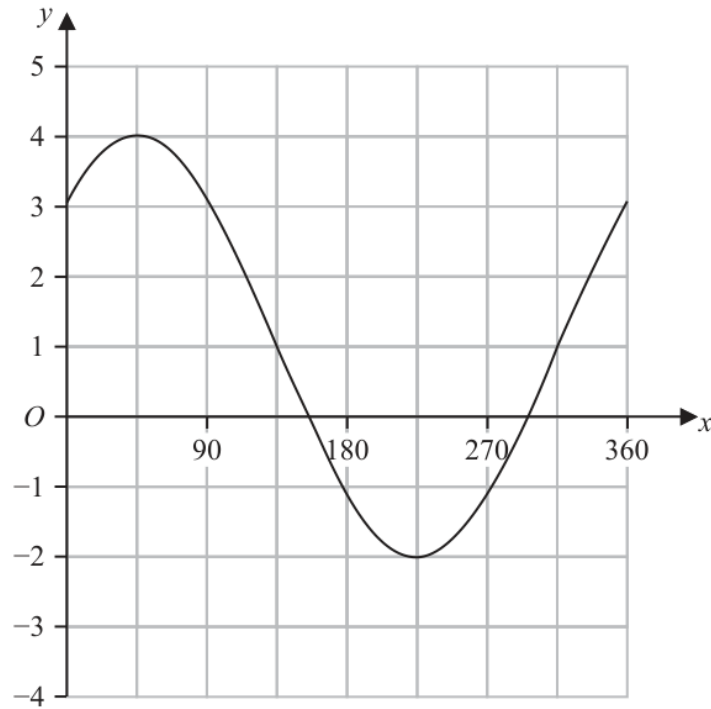
Note:

The period of the graph is still 360° (unchanged), so no horizontal stretch is applied.

PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Graphs of Functions**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

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May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

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FINAL ANSWER

$$a = 3, b = 45, c = 1$$

Note:

The period of the graph is still 360° (unchanged), so no horizontal stretch is applied.

PAST PAPER QUESTION**Q#: 24****Marks: 6**TOPIC: **Sequences**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

- 24** An arithmetic series P has first term a and common difference d
- The sum of the first 40 terms of P is 2070
- An arithmetic series Q has first term $2a$ and common difference $3d$
- The 25th term of Q is 186
- Work out the sum of the first 40 terms of Q
- Show clear algebraic working.

.....

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION**Q#: 24****Marks: 6****TOPIC: Sequences**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Required:

Series P : first term a , c.d. d ; $S_{40}(P) = 2070$. Series Q : first term $2a$, c.d. $3d$; 25th term of $Q = 186$. Find $S_{40}(Q)$.

Step:

Form equation from $S_{40}(P) = 2070$:

$$S_{40} = \frac{40}{2}[2a + 39d] = 20(2a + 39d) = 2070 \implies 2a + 39d = 103.5 \quad (1)$$

Step:

Form equation from 25th term of $Q = 186$:

$$T_{25}(Q) = 2a + 24(3d) = 2a + 72d = 186$$

Step:

Solve simultaneously – subtract (1) from (2):

$$33d = 82.5 \implies d = 2.5$$

Step:

Substituted $d = 2.5$ into (1) :

$$2a + 39(2.5) = 103.5 \implies 2a = 6 \implies a = 3$$

Step:

Compute $S_{40}(Q)$, with first term $= 2a = 6$ and c.d. $= 3d = 7.5$:

$$S_{40}(Q) = \frac{40}{2}[2(6) + 39(7.5)] = 20[12 + 292.5] = 20 \times 304.5 = 6090$$

FINAL ANSWER

$$S_{40}(Q) = 6090$$

PAST PAPER QUESTION**Q#: 24****Marks: 6**TOPIC: **Sequences**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

- 24** An arithmetic series P has first term a and common difference d
- The sum of the first 40 terms of P is 2070
- An arithmetic series Q has first term $2a$ and common difference $3d$
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(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION**Q#: 24****Marks: 6****TOPIC: Sequences**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Required:

Series P : first term a , c.d. d ; $S_{40}(P) = 2070$. Series Q : first term $2a$, c.d. $3d$; 25th term of $Q = 186$. Find $S_{40}(Q)$.

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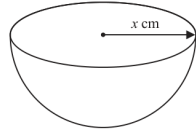
$$S_{40}(Q) = \frac{40}{2}[2(6) + 39(7.5)] = 20[12 + 292.5] = 20 \times 304.5 = 6090$$

FINAL ANSWER

$$S_{40}(Q) = 6090$$

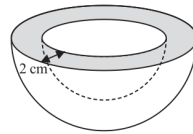
PAST PAPER QUESTION**Q#: 25****Marks: 5****TOPIC: Volume & Surface Area**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

25 The diagram shows a solid hemisphere, H Diagram **NOT**
accurately drawn

The radius of H is x cm
 The volume of H is 6174π cm³

A bowl is made by removing a solid hemisphere from H such that the uniform thickness of the bowl is 2 cm

Diagram **NOT**
accurately drawn

Work out the **total** surface area of the bowl.
 Give your answer in terms of π

..... cm²

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION**Q#: 25****Marks: 5****TOPIC: Volume & Surface Area**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***Required:**

Hemisphere H has radius x cm and volume 6174π cm³. A bowl is made by removing an inner hemisphere, leaving 2 cm thickness. Find total surface area in terms of π .

Step:

Find outer radius x . Volume of hemisphere = $\frac{2}{3}\pi r^3$:

$$\frac{2}{3}\pi x^3 = 6174\pi \implies x^3 = \frac{6174 \times 3}{2} = 9261 \implies x = \sqrt[3]{9261} = 21 \text{ cm}$$

Step:

Find inner radius:

$$r_{\text{inner}} = 21 - 2 = 19 \text{ cm}$$

Step:

Three surfaces to add:

1. Outer curved surface (half-sphere, radius 21):

$$A_1 = 2\pi(21)^2 = 2\pi \times 441 = 882\pi$$

2. Inner curved surface (half-sphere, radius 19):

$$A_2 = 2\pi(19)^2 = 2\pi \times 361 = 722\pi$$

3. Flat annular ring at the top rim:

$$A_3 = \pi(21)^2 - \pi(19)^2 = \pi(441 - 361) = 80\pi$$

Step:

Total surface area:

$$A_{\text{total}} = 882\pi + 722\pi + 80\pi = 1684\pi$$

FINAL ANSWER

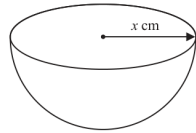
$$\text{Total surface area} = 1684\pi \text{ cm}^2$$

Note:

Three separate surfaces! Missing the flat annular ring is the most common error on this type of question.

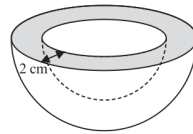
PAST PAPER QUESTION**Q#: 25****Marks: 5****TOPIC: Volume & Surface Area**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

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The radius of H is x cm
 The volume of H is 6174π cm³

A bowl is made by removing a solid hemisphere from H such that the uniform thickness of the bowl is 2 cm

Diagram **NOT**
accurately drawn

Work out the **total** surface area of the bowl.
 Give your answer in terms of π

..... cm²

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION**Q#: 25****Marks: 5****TOPIC: Volume & Surface Area**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***Required:**

Hemisphere H has radius x cm and volume 6174π cm³. A bowl is made by removing an inner hemisphere, leaving 2 cm thickness. Find total surface area in terms of π .

Step:

Find outer radius x . Volume of hemisphere = $\frac{2}{3}\pi r^3$:

$$\frac{2}{3}\pi x^3 = 6174\pi \implies x^3 = \frac{6174 \times 3}{2} = 9261 \implies x = \sqrt[3]{9261} = 21 \text{ cm}$$

Step:

Find inner radius:

$$r_{\text{inner}} = 21 - 2 = 19 \text{ cm}$$

Step:

Three surfaces to add:

1. Outer curved surface (half-sphere, radius 21):

$$A_1 = 2\pi(21)^2 = 2\pi \times 441 = 882\pi$$

2. Inner curved surface (half-sphere, radius 19):

$$A_2 = 2\pi(19)^2 = 2\pi \times 361 = 722\pi$$

3. Flat annular ring at the top rim:

$$A_3 = \pi(21)^2 - \pi(19)^2 = \pi(441 - 361) = 80\pi$$

Step:

Total surface area:

$$A_{\text{total}} = 882\pi + 722\pi + 80\pi = 1684\pi$$

FINAL ANSWER

$$\text{Total surface area} = 1684\pi \text{ cm}^2$$

Note:

Three separate surfaces! Missing the flat annular ring is the most common error on this type of question.